

Experiment-9

To Implement Travelling Salesman Problem

Aim

To develop a Python program to solve the Travelling Salesman Problem (TSP) using the Brute Force approach.

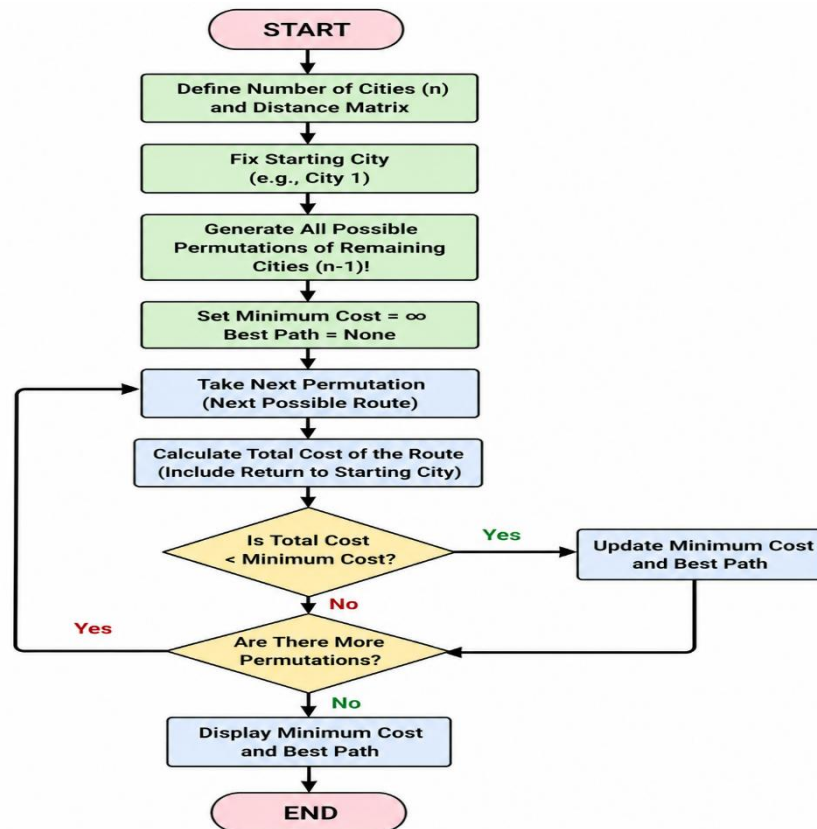
Objective

- To implement the Travelling Salesman Problem (TSP).
- To find the minimum cost path that visits every city exactly once and returns to the starting city.
- To understand optimization techniques in Artificial Intelligence.

Algorithm

1. Start the program.
2. Define the distance matrix for all cities.
3. Generate all possible routes using permutations.
4. Calculate the total cost of each route.
5. Compare the route cost with the minimum cost.
6. Update the minimum cost if a shorter route is found.
7. Display the minimum travel cost.
8. Stop the program.

Flowchart:



Code:

```
from itertools import permutations
```

```
g = [[0,10,15,20],  
     [10,0,35,25],  
     [15,35,0,30],  
     [20,25,30,0]]
```

```
n = 4
```

```
cost = 9999
```

```
for p in permutations(range(1, n)):
```

```
    c = g[0][p[0]]
```

```
    for i in range(len(p)-1):
```

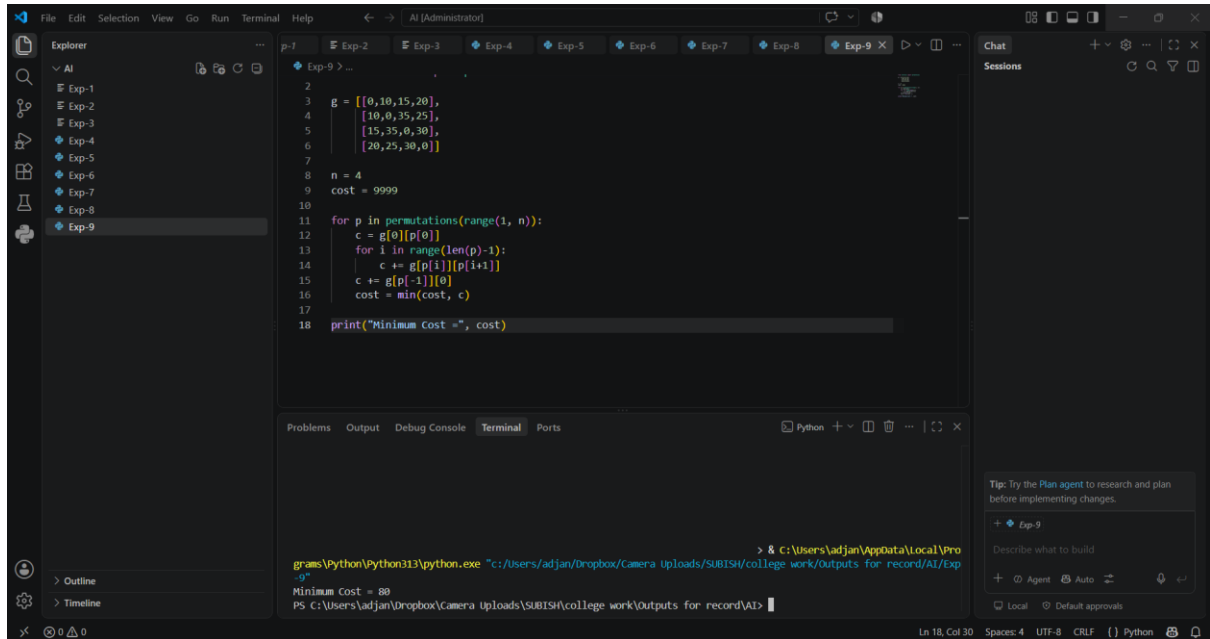
```
        c += g[p[i]][p[i+1]]
```

```
c += g[p[-1]][0]
```

```
cost = min(cost, c)
```

```
print("Minimum Cost =", cost)
```

Output:



The screenshot shows a VS Code editor with a Python script for solving the Travelling Salesman Problem (TSP). The script defines a cost matrix `g` and a function to calculate the minimum cost by evaluating all possible routes. The output in the terminal shows the minimum cost is 80.

```
2  
3 g = [[0,10,15,20],  
4       [10,0,35,25],  
5       [15,35,0,30],  
6       [20,25,30,0]]  
7  
8 n = 4  
9 cost = 9999  
10  
11 for p in permutations(range(1, n)):  
12     c = g[0][p[0]]  
13     for i in range(len(p)-1):  
14         c += g[p[i]][p[i+1]]  
15     c += g[p[-1]][0]  
16     cost = min(cost, c)  
17  
18 print("Minimum Cost =", cost)
```

Terminal Output:

```
> & C:\Users\adjan\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/adjan/Dropbox/Camera Uploads/SUBISH/college work/Outputs for record/AI/Exp-9"  
Minimum Cost = 80  
PS C:\Users\adjan\Dropbox\Camera Uploads\SUBISH\college work\Outputs for record\AI>
```

Result:

The Travelling Salesman Problem (TSP) was successfully solved, and the minimum travel cost was obtained.

Conclusion:

The Travelling Salesman Problem was solved by evaluating all possible routes and selecting the one with the minimum travel cost.